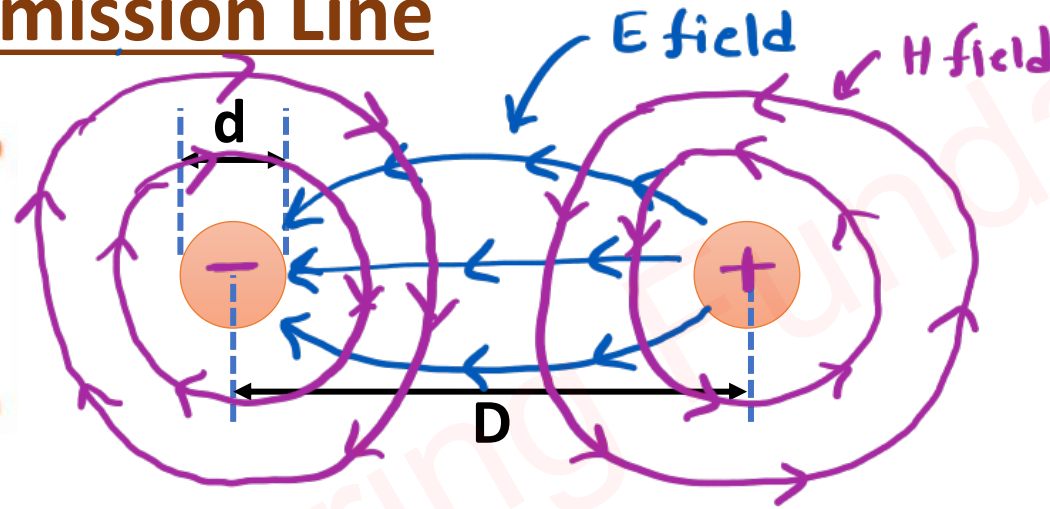
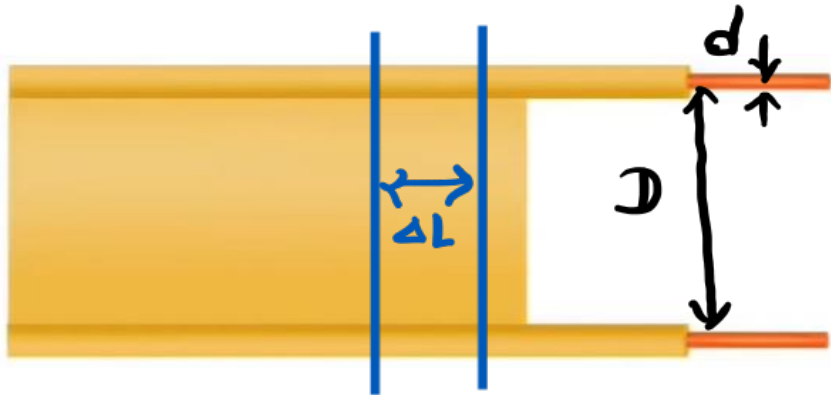


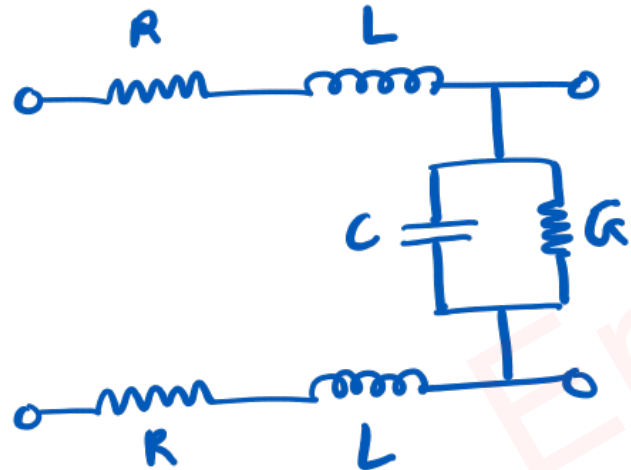
Transmission Lines of Microwave

Two Parallel Wire Transmission Line



$$L = \frac{\mu}{\pi} \ln \left[\frac{2D}{d} \right]$$

$$C = \frac{\pi \epsilon}{\ln \left[\frac{2D}{d} \right]}$$



→ Characteristics Impedance

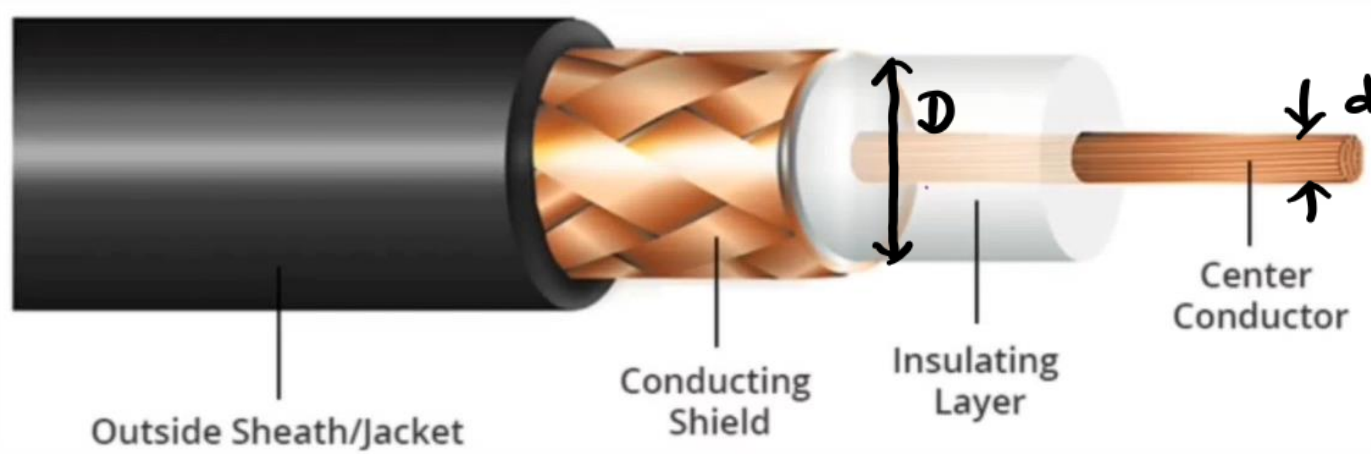
$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

→ For Lossless T_x line

$$R = G = 0$$

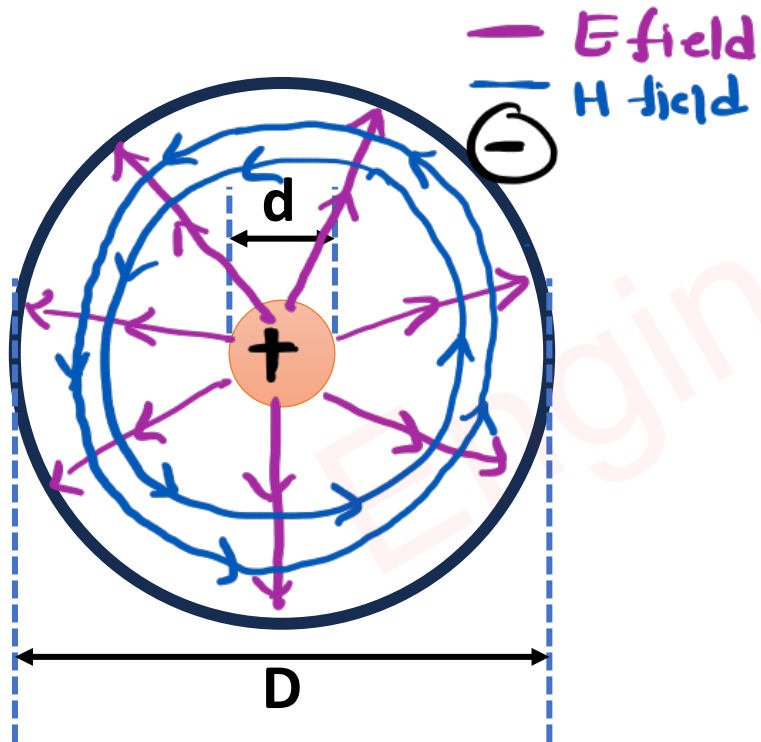
$$Z_0 = \sqrt{L/C}$$

Coaxial Transmission Line



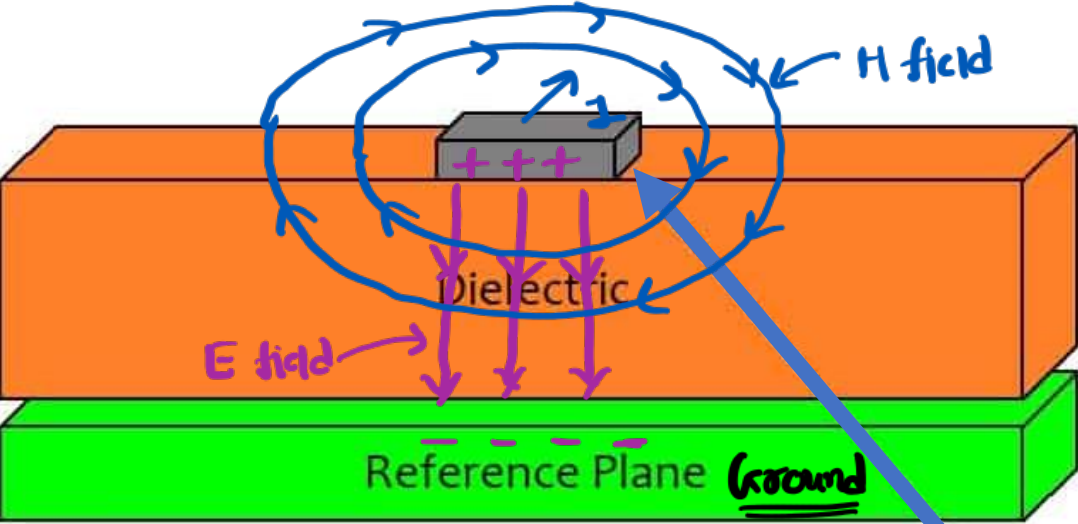
$$L = \frac{\mu}{2\pi} \ln \left[\frac{D}{d} \right]$$

$$C = \frac{2\pi\epsilon}{\ln \left[\frac{D}{d} \right]}$$

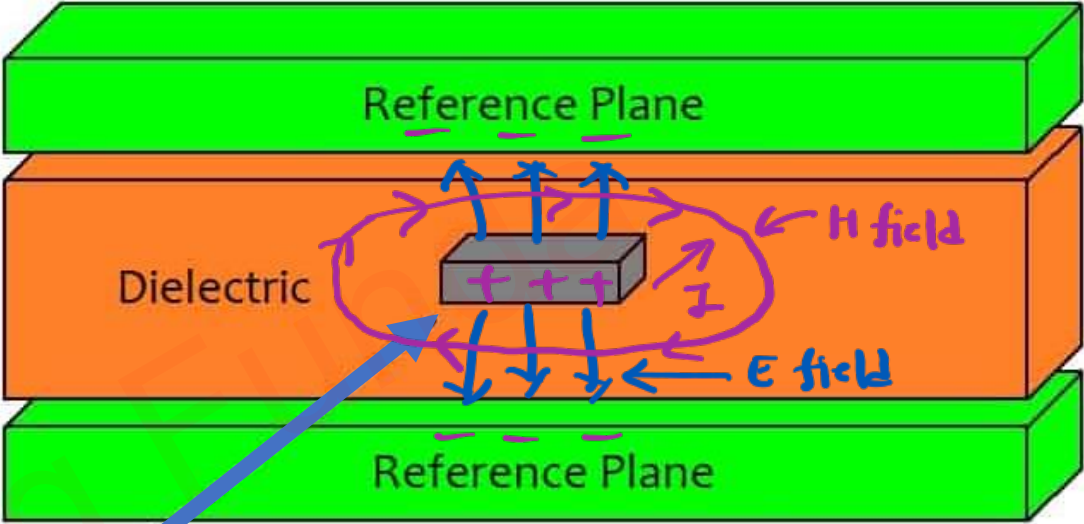


$$\rightarrow Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Microstrip Transmission Line



[Microstrip Transmission Line]

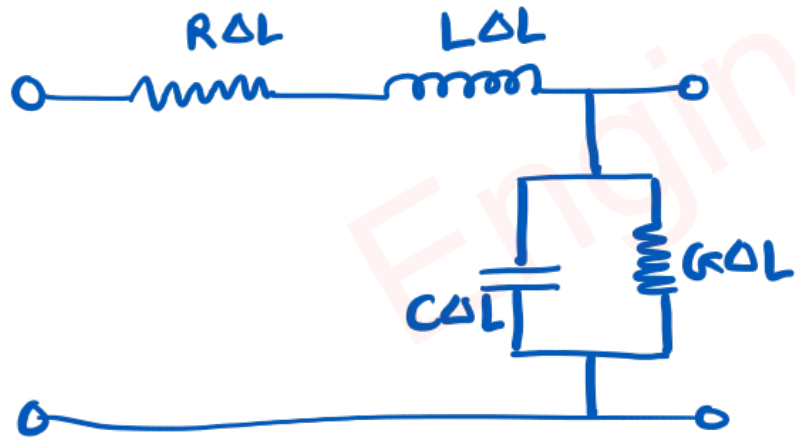
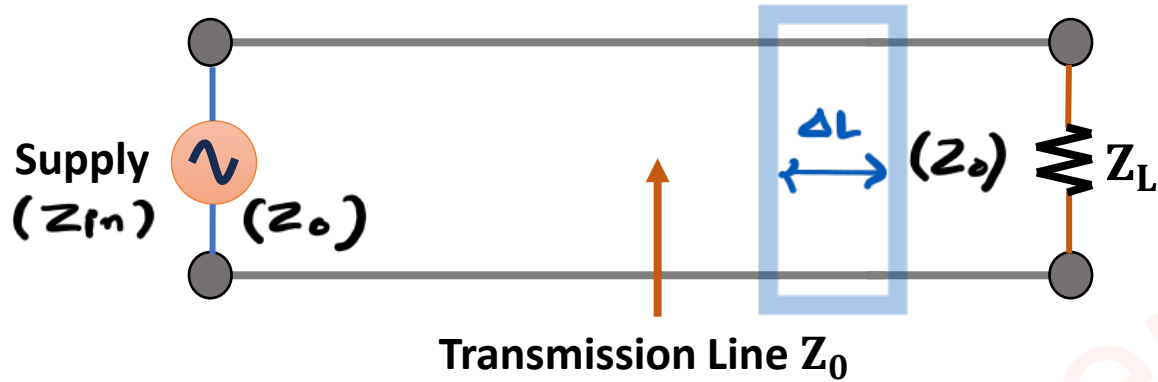


[Stripline Transmission Line]

High-speed transmission line

Equivalent Circuit of Transmission Line

□ Two Parallel Metal is Transmission Line.



⇒ R & G are Lossy components of Tx line.

Characteristic Impedance of Transmission Line

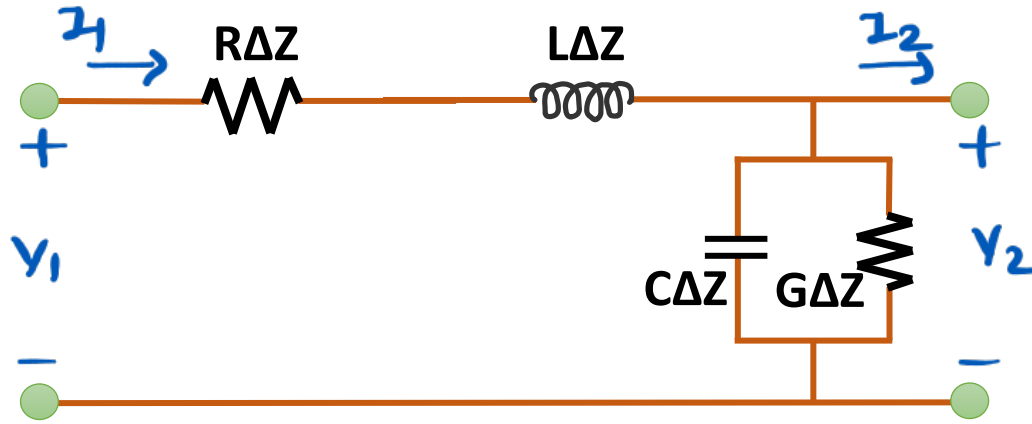
- Characteristic Impedance is the Impedance of the Transmission Line measured at I/P or O/P, Provided the Length of the Transmission Line is infinite.

$$\rightarrow Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

⇒ For lossless Tx line. $[R = G = 0]$

$$\rightarrow Z_0 = \sqrt{L/C}$$

Transmission Lines Equations



⇒ As per KVL, Voltage drop Across Tx Line

$$\Rightarrow \Delta V = V_2 - V_1 = - \left[R\Delta z I_1 + L\Delta z \frac{dI_1}{dt} \right]$$

$$\Rightarrow \frac{\Delta V}{\Delta z} = - \left[R + L \frac{d}{dt} \right] I_1 \quad \text{--- (1)}$$

⇒ As per KCL, Current drop Across Tx Line

$$\Rightarrow \Delta I = I_2 - I_1 = - \left[G\Delta z V_2 + C\Delta z \frac{dV_2}{dt} \right]$$

$$\Rightarrow \frac{\Delta I}{\Delta z} = - \left[G + C \frac{d}{dt} \right] V_2 \quad \text{--- (2)}$$

~> Write eqⁿ ① & ② in form of V & I .

$$\Rightarrow \frac{dV}{dz} = -[R + L \frac{d}{dt}] I \quad \text{--- ③}$$

$$\Rightarrow \frac{dI}{dz} = -[G + C \frac{d}{dt}] V \quad \text{--- ④}$$

~> Here, $\frac{d}{dt} = j\omega$

$$\Rightarrow \frac{dV}{dz} = -[R + j\omega L] I \quad \text{--- ⑤}$$

$$\Rightarrow \frac{dI}{dz} = -[G + j\omega C] V \quad \text{--- ⑥}$$

~> To get Tx line eqⁿ differentiate eqⁿ ⑤ w.r.t. z

$$\Rightarrow \frac{d^2 V}{dz^2} = -[R + j\omega L] \frac{dI}{dz}$$

$$\Rightarrow \frac{d^2 V}{dz^2} = [R + j\omega L][G + j\omega C] V \quad \text{--- ⑦}$$

∴ eqn (A) T_x line eqn and it is similar to wave eqn

$$\Rightarrow \frac{d^2 V}{dz^2} = \gamma^2 V \quad \text{--- (F)}$$

∴ γ = propagation constant

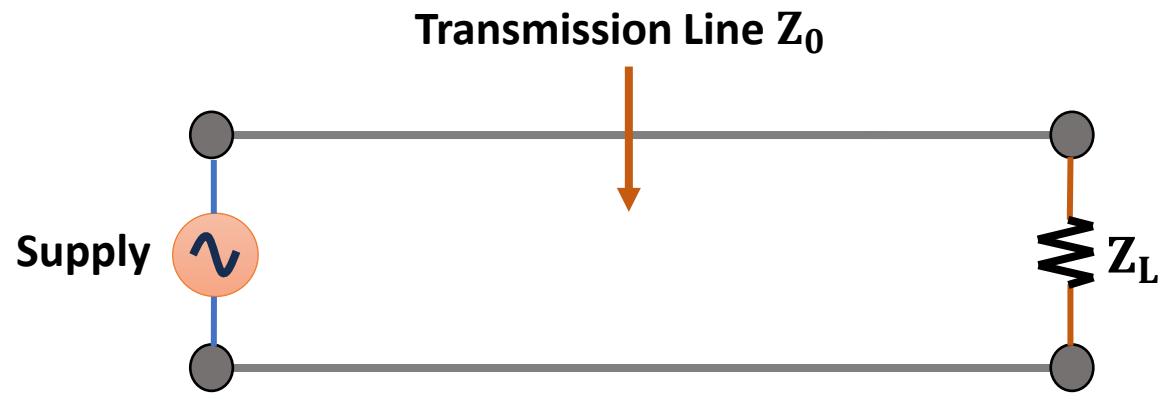
$$= \sqrt{(R + j\omega L)(G + j\omega C)}$$

∴ Sol.ⁿ of eqn (F) is

$$\Rightarrow V = \underbrace{V_1 e^{-\gamma z}}_{\substack{\uparrow \\ \text{Incident} \\ \text{Signal}}} + \underbrace{V_2 e^{+\gamma z}}_{\substack{\uparrow \\ \text{Reflected} \\ \text{Signal.}}}$$

∴ Similarly,

$$\Rightarrow I = \underbrace{I_1 e^{-\gamma z}}_{\substack{\uparrow \\ \text{Incident} \\ \text{Signal}}} + \underbrace{I_2 e^{+\gamma z}}_{\substack{\uparrow \\ \text{Reflected} \\ \text{Signal}}}$$



Incident
Signal
 $[V_1 e^{-\gamma z}]$

Reflected
Signal
 $[V_2 e^{+\gamma z}]$

Propagation Constant, Attenuation Constant, & Phase Constant of the Transmission Line

→ From Tx line wave eqⁿ

$$\Rightarrow \gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$= \underline{\alpha} + j\underline{\beta}$$

Attenuation
Constant

Phase
Constant

→ For lossless Tx line, $[R = G = 0]$

$$\Rightarrow \gamma = \sqrt{(0 + j\omega L)(0 + j\omega C)}$$

$$\Rightarrow \gamma = j\omega\sqrt{LC}$$

$$\Rightarrow \alpha = 0, \quad \beta = \omega\sqrt{LC}$$

→ α and β

$$\Rightarrow \alpha = \frac{1}{2} \left[G\sqrt{\frac{L}{C}} + R\sqrt{\frac{C}{L}} \right]$$

$$\Rightarrow \beta = \omega\sqrt{LC} \left[1 + \frac{1}{8} \left(\frac{R}{\omega L} - \frac{G}{\omega C} \right)^2 \right]$$

Characteristics Impedance

□ Characteristic Impedance Z_0 is the Impedance of the Transmission Line measured at I/P or O/P, Provided the Length of the Transmission Line is infinite.

⇒ Sol.ⁿ of wave eq.ⁿ

$$\Rightarrow V = V_1 e^{-\gamma z} + V_2 e^{+\gamma z}$$

⇒ Tx line eq.ⁿ

$$\Rightarrow \frac{dV}{dz} = -(R + j\omega L) I$$

$$\Rightarrow I = -\frac{1}{R + j\omega L} \left(\frac{dV}{dz} \right)$$

$$\Rightarrow I = \frac{-1}{R + j\omega L} \frac{d(V_1 e^{-\gamma z} + V_2 e^{+\gamma z})}{dz}$$

$$\Rightarrow I = \frac{-1}{R + j\omega L} (V_1 (-\gamma) e^{-\gamma z} + V_2 (\gamma) e^{+\gamma z})$$

$$\Rightarrow I = \frac{\gamma}{R + j\omega L} (V_1 e^{-\gamma z} - V_2 e^{\gamma z})$$

$$\Rightarrow \text{[Hence, } \gamma = \sqrt{(R + j\omega L)(G + j\omega C)} \text{]}$$

$$\Rightarrow I = \frac{\sqrt{(R + j\omega L)(G + j\omega C)}}{R + j\omega L} (V_1 e^{-\gamma z} - V_2 e^{\gamma z})$$

$$\Rightarrow I = \sqrt{\frac{G + j\omega C}{R + j\omega L}} (V_1 e^{-\gamma z} - V_2 e^{\gamma z})$$

$$\Rightarrow I = \frac{1}{Z_0} (V_1 e^{-\gamma z} - V_2 e^{\gamma z})$$

$\Rightarrow$ Hence, Characteristic Impedance Z_0 is

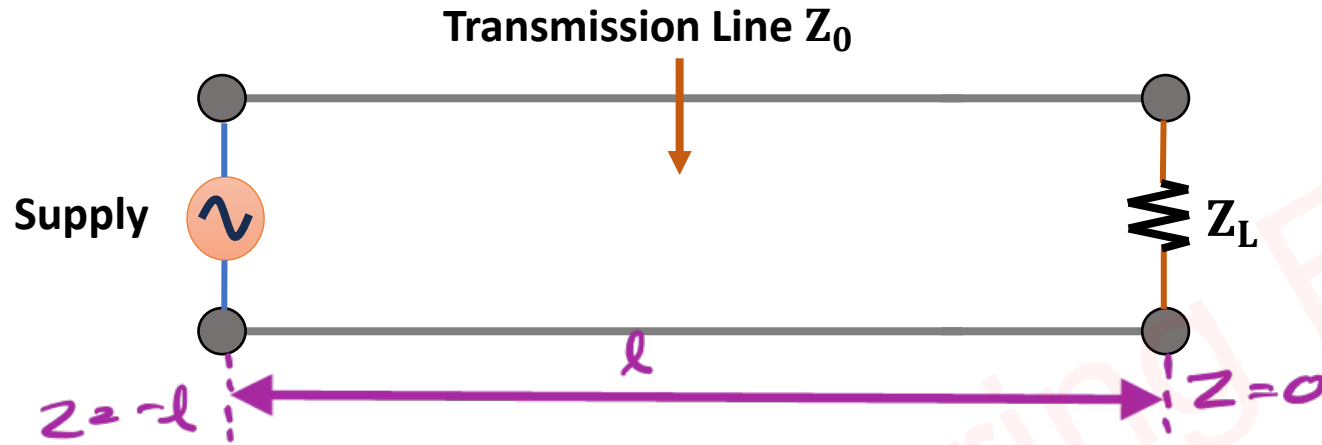
$$\Rightarrow Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$\Rightarrow$ For lossless T_x line,
[$R = G = 0$]

$$\Rightarrow Z_0 = \sqrt{L/C}$$

Reflection Coefficient

□ Reflection Coefficient explains the amount of reflection of the signal from the load.



~> Based on Voltage & Current eq.ⁿ of Tx line

$$\Rightarrow V = \underbrace{V_1 e^{-\gamma z}}_{\text{Incident Signal}} + \underbrace{V_2 e^{\gamma z}}_{\text{Reflected Signal}}$$

$$\Rightarrow I = \underbrace{I_1 e^{-\gamma z}}_{\text{Incident Signal}} + \underbrace{I_2 e^{\gamma z}}_{\text{Reflected Signal}}$$

~> So, above eq.ⁿ are :

$$\Rightarrow V = V_i e^{-\gamma z} + V_r e^{\gamma z}$$

$$\Rightarrow I = I_i e^{-\gamma z} + I_r e^{\gamma z}$$

↪ At load $z = 0$

$$\Rightarrow V_L = V_i + V_r \quad \Rightarrow I_L = I_i + I_r$$

↪ At Source $z = -l$

$$\Rightarrow V_S = V_i e^{\gamma l} + V_r e^{-\gamma l} \quad \Rightarrow I_S = I_i e^{\gamma l} + I_r e^{-\gamma l}$$

↪ Characteristic Impedance Z_0

$$\Rightarrow Z_0 = \frac{V_i}{I_i} = -\frac{V_r}{I_r}$$

↪ Reflection Coefficient γ

$$\Rightarrow \gamma = \frac{V_r}{V_i} = -\frac{I_r}{I_i}$$

↪ Load Impedance Z_L

$$\Rightarrow Z_L = \frac{V_L}{I_L} = \frac{V_i + V_r}{I_i + I_r}$$

$$\Rightarrow Z_L = \frac{V_i}{I_i} \left(\frac{1 + (\sqrt{r}/V_i)}{1 + (I_0/I_i)} \right)$$

$$\Rightarrow Z_L = Z_0 \left(\frac{1+r}{1-r} \right)$$

$$\Rightarrow Z_L(1-r) = Z_0(1+r)$$

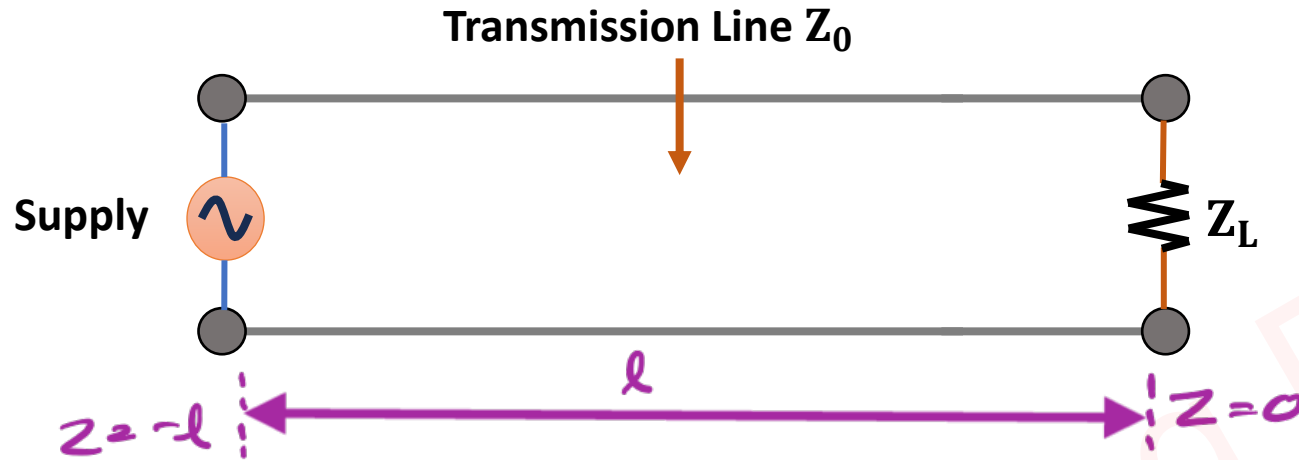
$$\Rightarrow Z_L - Z_0 = r(Z_L + Z_0)$$

$$\Rightarrow \boxed{r = \frac{Z_L - Z_0}{Z_L + Z_0}}$$

∴ For perfect Impedance matching $Z_L = Z_0$

$$\Rightarrow r = \frac{0}{2Z_L} = 0$$

Input Impedance



~> Based on Voltage & Current eq.ⁿ of Tx line

$$\Rightarrow V = \underbrace{V_1 e^{-\gamma z}}_{\text{Incident Signal}} + \underbrace{V_2 e^{\gamma z}}_{\text{Reflected Signal}}$$

$$\Rightarrow I = \underbrace{I_1 e^{-\gamma z}}_{\text{Incident Signal}} + \underbrace{I_2 e^{\gamma z}}_{\text{Reflected Signal}}$$

~> So, above eq.ⁿ are :

$$\Rightarrow V = V_i e^{-\gamma z} + V_r e^{\gamma z}$$

$$\Rightarrow I = I_i e^{-\gamma z} + I_r e^{\gamma z}$$

~> At load $z = 0$

$$\Rightarrow V_L = V_i + V_r$$

$$\Rightarrow I_L = I_i + I_r$$

↪ At Source $z = -l$

$$\Rightarrow V_s = V_i e^{\gamma l} + V_r e^{-\gamma l} \quad \Rightarrow I_s = I_i e^{\gamma l} + I_r e^{-\gamma l}$$

↪ Characteristic Impedance Z_0

$$\Rightarrow Z_0 = \frac{V_i}{I_i} = -\frac{V_r}{I_r}$$

↪ Reflection Coefficient γ

$$\Rightarrow \gamma = \frac{V_r}{V_i} = -\frac{I_r}{I_i}$$

↪ Input Impedance Z_{in}

$$\Rightarrow Z_{in} = \frac{V_s}{I_s} = \frac{V_i e^{\gamma l} + V_r e^{-\gamma l}}{I_i e^{\gamma l} + I_r e^{-\gamma l}} = \frac{V_i (e^{\gamma l} + (V_r/V_i) e^{-\gamma l})}{I_i (e^{\gamma l} + (I_r/I_i) e^{-\gamma l})}$$

$$\Rightarrow Z_{in} = Z_0 \left[\frac{e^{\gamma l} + \gamma e^{-\gamma l}}{e^{\gamma l} - \gamma e^{-\gamma l}} \right]$$

$$\left[\text{Reflection Coefficient } \gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \right]$$

$$\Rightarrow Z_{in} = Z_0 \left[\frac{e^{\gamma l} + \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-\gamma l}}{e^{\gamma l} - \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-\gamma l}} \right]$$

$$\Rightarrow Z_{in} = Z_0 \left[\frac{e^{\gamma l} (Z_L + Z_0) + (Z_L - Z_0) e^{-\gamma l}}{e^{\gamma l} (Z_L + Z_0) - (Z_L - Z_0) e^{-\gamma l}} \right]$$

$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_L (e^{\gamma l} + e^{-\gamma l}) + Z_0 (e^{\gamma l} - e^{-\gamma l})}{Z_L (e^{\gamma l} - e^{-\gamma l}) + Z_0 (e^{\gamma l} + e^{-\gamma l})} \right]$$

$$\left[\text{For lossless Tx line } \alpha = 0, \gamma = \alpha + j\beta = j\beta \right]$$

$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_L (e^{j\beta l} + e^{-j\beta l}) + Z_0 (e^{j\beta l} - e^{-j\beta l})}{Z_L (e^{j\beta l} - e^{-j\beta l}) + Z_0 (e^{j\beta l} + e^{-j\beta l})} \right]$$

$$\left[\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}, \quad \cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2} \right]$$

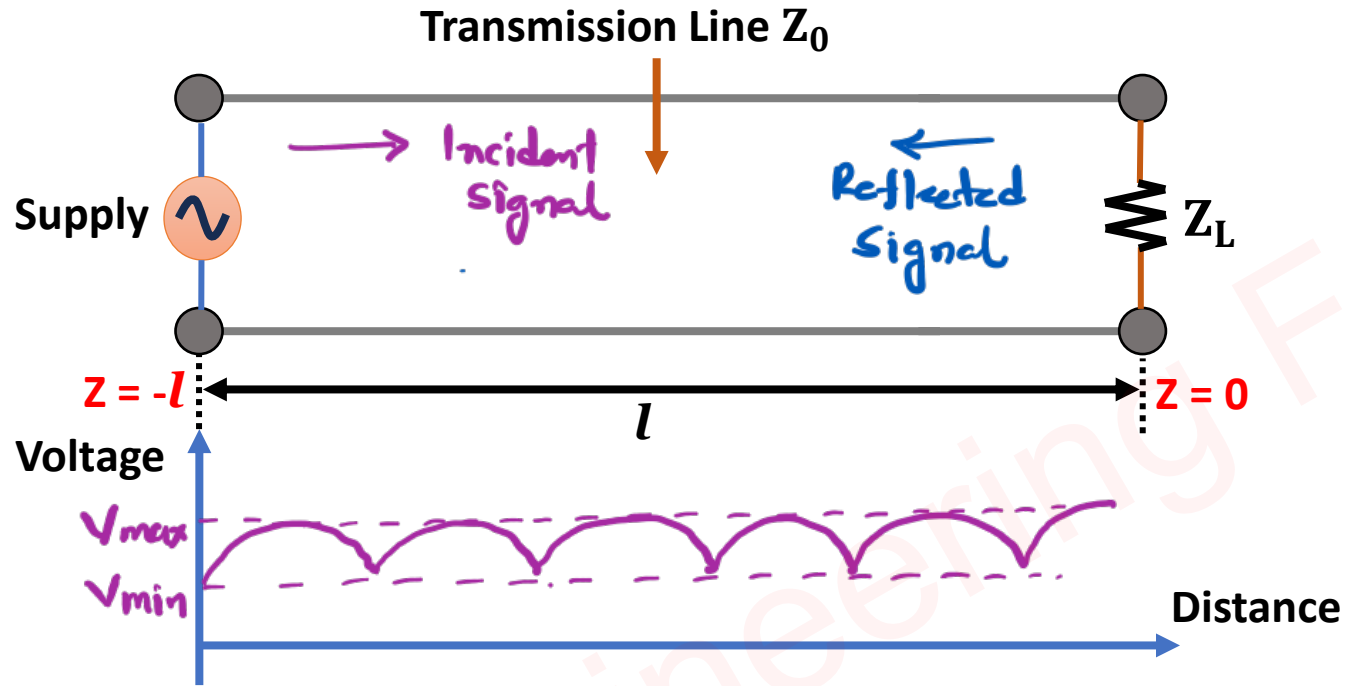
$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_L (1 \cos \beta l) + Z_0 (j \sin \beta l)}{Z_L (j \sin \beta l) + Z_0 (1 \cos \beta l)} \right]$$

$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_L + j Z_0 \tan \beta l}{Z_0 + j Z_L \tan \beta l} \right]$$

$$\left[\text{Normalized Impedance, } \bar{Z}_{in} = \frac{Z_{in}}{Z_0}, \quad \bar{Z}_L = \frac{Z_L}{Z_0} \right]$$

$$\Rightarrow \bar{Z}_{in} = \frac{\bar{Z}_L + j \tan \beta l}{1 + j \bar{Z}_L \tan \beta l}$$

VSWR, Z_{MAX} and Z_{MIN} of Tx Line



→ Voltage eq.ⁿ of Tx line

$$\Rightarrow V = \underbrace{V_i e^{-\gamma z}}_{\text{Incident Signal}} + \underbrace{V_r e^{+\gamma z}}_{\text{Reflected Signal.}}$$

→ Max. Voltage V_{max}

$$\Rightarrow V_{max} = V_i + V_R$$

→ Min. Voltage V_{min}

$$\Rightarrow V_{min} = V_i - V_R$$

→ Voltage Standing Wave Ratio (VSWR)

$$\Rightarrow VSWR = \frac{V_{max}}{V_{min}} = \frac{V_i + V_R}{V_i - V_R}$$

$$\Rightarrow VSWR = \frac{V_i (1 + V_R/V_i)}{V_i (1 - V_R/V_i)}$$

$$\Rightarrow \boxed{VSWR = \frac{1 + |r|}{1 - |r|}}$$

→ Range of Reflection Coefficient is from 0 to 1.

Range of VSWR is from 1 to ∞ .

→ Reflection Coefficient, $r = \frac{Z_L - Z_0}{Z_L + Z_0}$

$$\leadsto \text{If } Z_L > Z_0 \Rightarrow \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \Rightarrow VSWR = \frac{Z_L}{Z_0}$$

$$\leadsto \text{If } Z_0 > Z_L \Rightarrow \Gamma = \frac{Z_0 - Z_L}{Z_L + Z_0} \Rightarrow VSWR = \frac{Z_0}{Z_L}$$

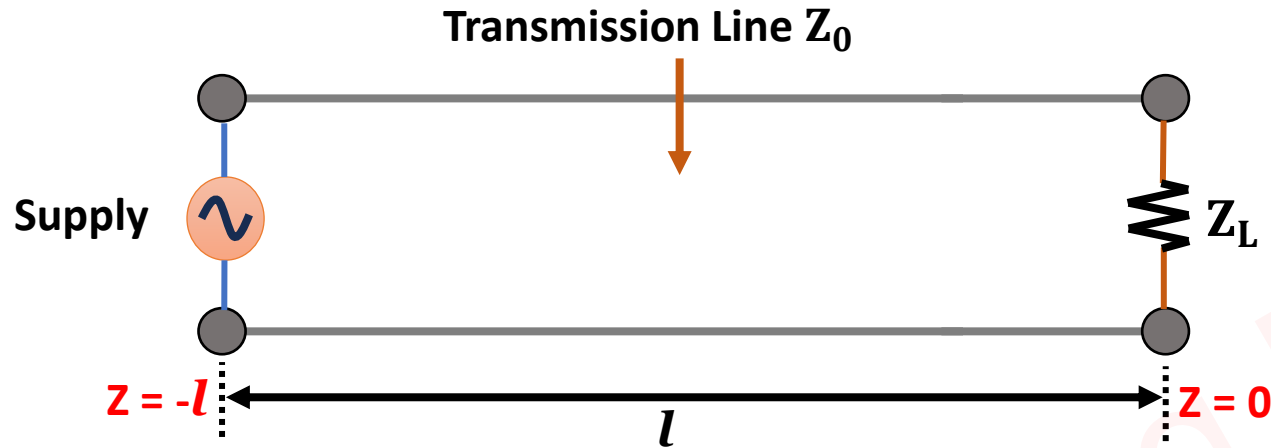
$\leadsto$ Max. Impedance of T_x line.

$$\begin{aligned} \Rightarrow Z_{\max} &= \frac{V_{\max}}{I_{\min}} \\ &= \frac{V_{\max}}{V_{\min}} \times \frac{V_{\min}}{I_{\min}} \\ &= VSWR \times Z_0 \end{aligned}$$

$\leadsto$ Min. Impedance of T_x line

$$\begin{aligned} \Rightarrow Z_{\min} &= \frac{V_{\min}}{I_{\max}} \\ &= \frac{V_{\min}}{V_{\max}} \times \frac{V_{\max}}{I_{\max}} \\ &= \frac{1}{VSWR} \times Z_0 \\ &= \frac{Z_0}{VSWR} \end{aligned}$$

Transmittance Coefficient



~ Voltage eq.ⁿ of Tx line

$$\Rightarrow V = \underbrace{V_i e^{-\gamma z}}_{\text{Incident Signal}} + \underbrace{V_r e^{\gamma z}}_{\text{Reflected Signal}}$$

~ At load, $z = 0$

$$\Rightarrow V_L = V_i + V_r$$

→ Transmitted signal

$$\Rightarrow V_t = V_i + V_r$$

→ Incident signal

$$V_i = V_i$$

→ Transmittance coefficient $T = \frac{\text{Transmitted signal}}{\text{Incident signal}}$

$$= \frac{V_t}{V_i} = \frac{V_i + V_r}{V_i}$$

$$= 1 + \left(\frac{V_r}{V_i} \right)$$

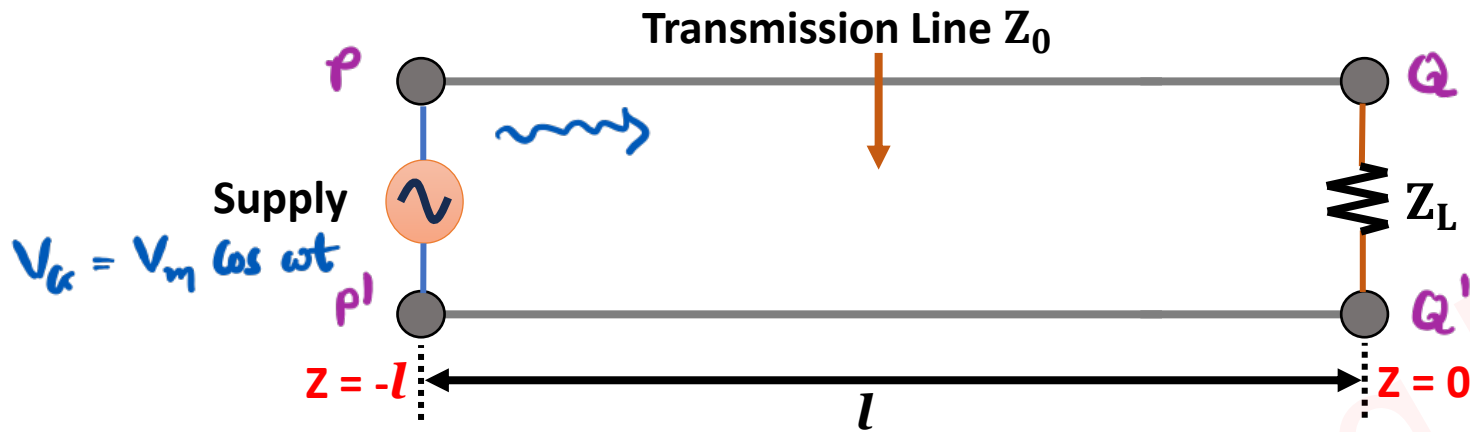
$$\boxed{T = 1 + r}$$

→ [Reflection coefficient $r = \frac{Z_L - Z_0}{Z_L + Z_0}$]

$$\Rightarrow T = 1 + \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_L + \cancel{Z_0} + Z_L - \cancel{Z_0}}{Z_L + Z_0}$$

$$\boxed{\Rightarrow T = \frac{2Z_L}{Z_L + Z_0}}$$

Generation of Phase Delay



⇒ Let's assume 100% Impedance matching of Supply & Tx line.

$$\Rightarrow V_{PP'} = V_m \cos \omega t$$

⇒ Let's assume 100% Impedance matching of Tx line & load.

$$\Rightarrow V_{QQ'} = V_m \cos \omega(t + \Delta t)$$

$$\left[v_p = \frac{L}{\Delta t} \Rightarrow \Delta t = \frac{L}{v_p} \right]$$

$$\Rightarrow V_{QQ'} = V_m \cos \left(\omega t + \frac{\omega L}{v_p} \right)$$

$$\Rightarrow V_{QQ'} = V_m \cos(\omega t + \phi)$$

$$\Rightarrow \text{Hence, phase } \phi = \frac{\omega L}{V_p}$$

$$\Rightarrow f = 1 \text{ kHz}, L = 1 \text{ m}, V_p = 2 \times 10^8 \text{ m/sec}$$

$$\phi = \frac{2\pi f L}{V_p} = \frac{2\pi \times 10^3 \times 1}{2 \times 10^8} = 3.14 \times 10^{-5} \text{ rad}$$

$$\Rightarrow f = 1 \text{ kHz}, L = 1 \text{ m}, V_p = 2 \times 10^8 \text{ m/sec}$$

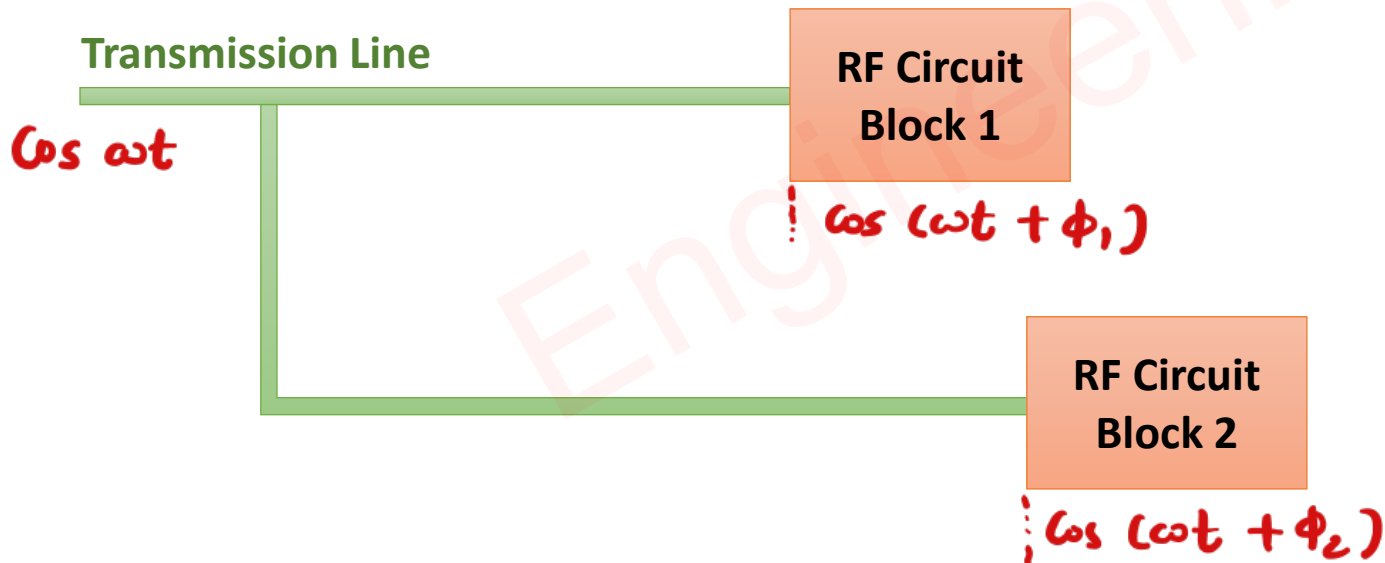
$$\phi = \frac{2\pi f L}{V_p} = \frac{2\pi \times 10^9 \times 1}{2 \times 10^8} = 31.4 \text{ rad}$$

Microwave Effects on Tx Line

□ Phase Delay

- Phase Delay happens due to the length of the Transmission Line.
- Phase Delay generates synchronization issues in the RF Circuits.
- Phase Delay also depends on the Frequency of Microwave Signals.

$$\text{Phase } \phi = \frac{2\pi fL}{v_p}$$



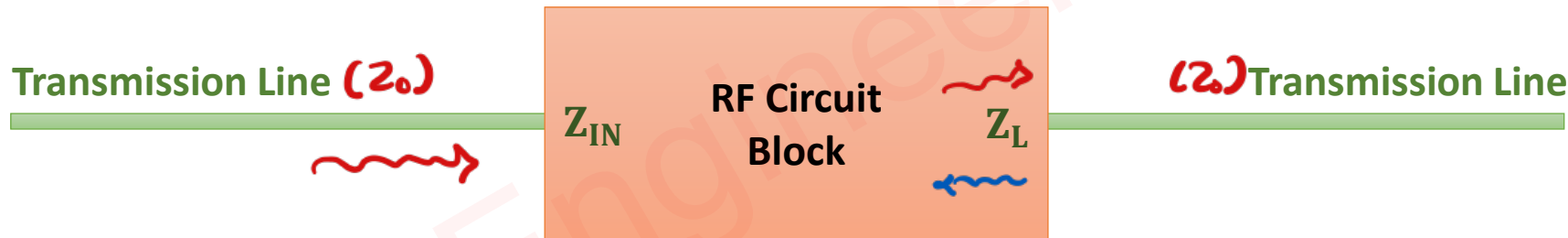
□ Reflection of Signals

- Reflection of Signals happens due to the Impedance mismatch.

$$\text{Reflection Coefficient } \Upsilon = \frac{Z_L - Z_0}{Z_L + Z_0}$$

- Characteristics Impedance Z_0 depends on the operating frequency.

$$\text{Characteristics Impedance } Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

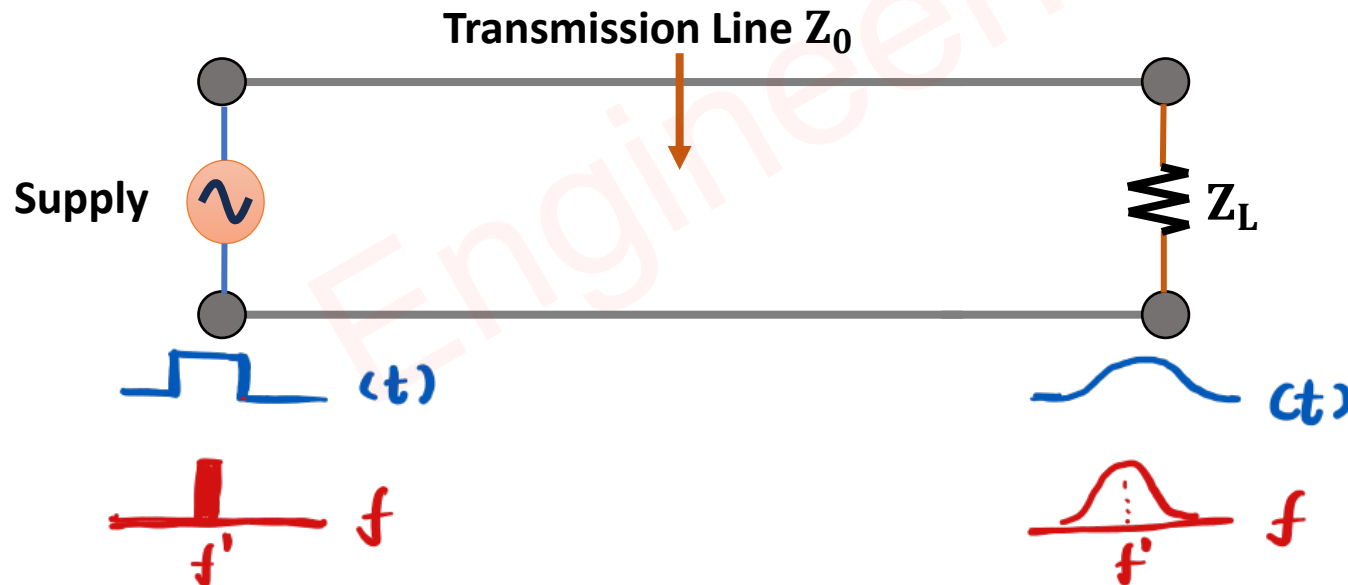


❑ Power Loss

- Power Loss in Transmission Lines happens due to Conducting and Insulating Materials.
- In Insulating Materials, Microwave signal attenuation happens.
- In Conducting Materials, Microwave signals have issues due to skin effects and signal absorption.
- Power Loss increases with an increase in the frequency of Microwave Signal.

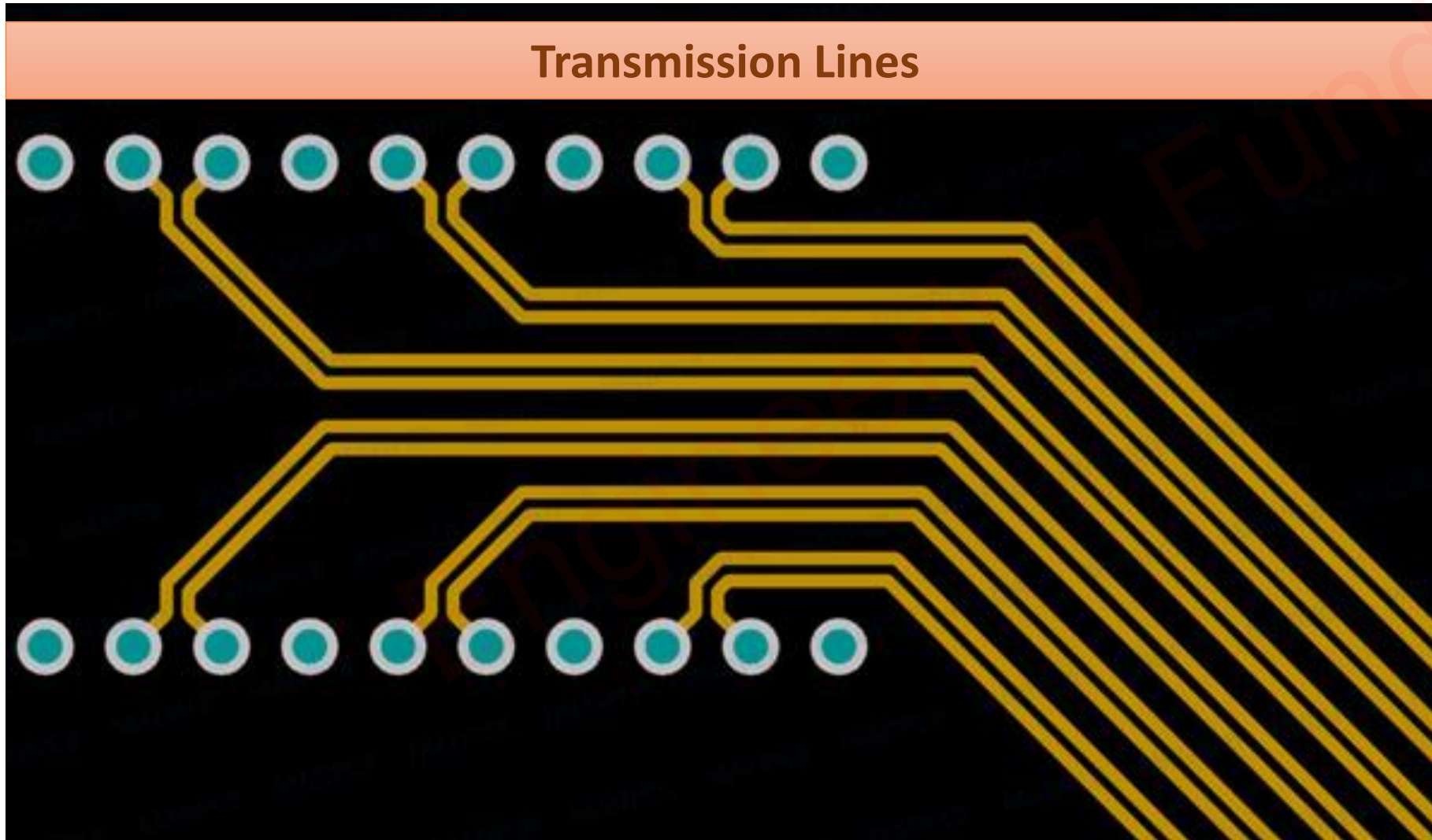
❑ Signal Dispersion

- Signal Dispersion results in the broadening of transmitted pulses.
- Due to the broadening of pulses, signals have more frequency components.
- Signal Dispersion also depends on the frequency and the Length of the Transmission Line.



❑ Coupling Loss

- EM signal couples in between transmission lines and RF Circuits.
- Coupling Loss also depends on the spacing between transmission lines and Isolation between Transmission Lines.



Limitations of Tx Line

❑ Separation Between Two Parallel wires.

- At Lower Frequencies, the wavelengths will be higher, Hence we need a large separation between two parallel wires.
- At Higher Frequencies, the wavelengths will be shorter, Hence we need a small separation between two parallel wires.
- So, After the fabrication of the Transmission Line, we can not change the operating frequency of the Transmission Line.

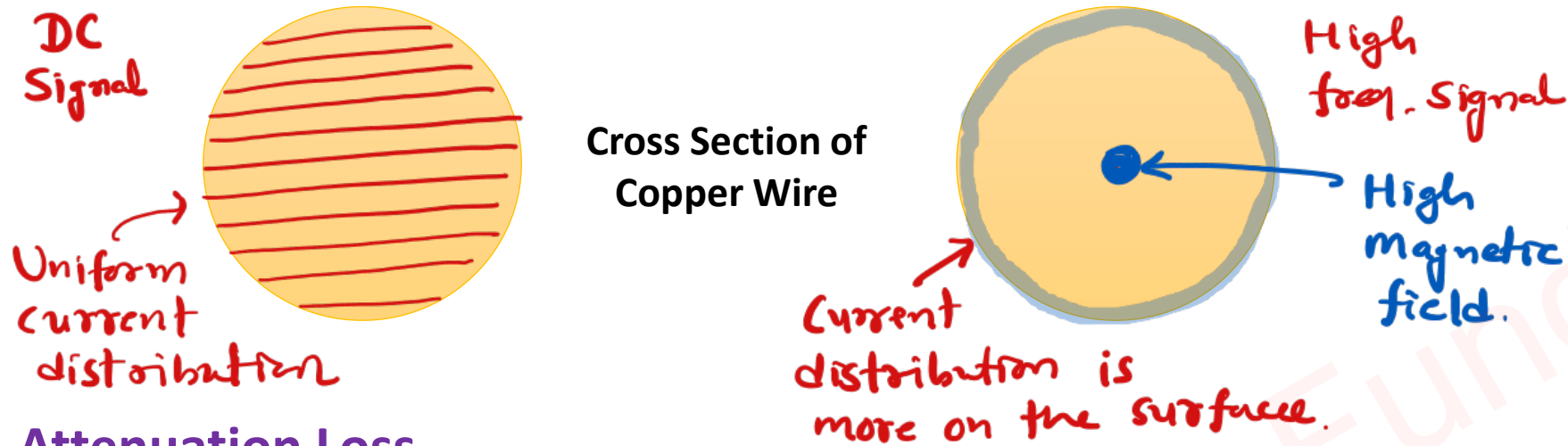
$$Z_o = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

❑ Copper Loss.

- Copper Loss depends on the Surface area of the Metal (Copper).
- The lower the surface area, the higher the Copper Loss with the Transmission Line.

❑ Skin Effect Loss.

- At Microwave frequencies, the skin effect creates losses of signal.
- At Microwave frequencies, Current distribution happens on the perimeter of the wire, which means the current distribution repeats towards the surface of the wire.
- If we further increase the frequency, signal is getting radiated in the free space.
- So, at higher frequencies, loss due to skin effect is higher.



□ Attenuation Loss.

- At Microwave frequencies, the wavelength is shorter, so the separation is also short between two wires, which leads to higher attenuation of signal.
- At Microwave frequencies, due to attenuation of signal, if we increase the power, there is a possibilities of dielectric breakdown.

□ Radiation and Induction Loss.

- At Microwave frequencies, the wavelength is shorter, so it may radiate EM waves.
- At Microwave frequencies, Lumped parameters like inductance is included, which results in induction loss.

□ Power Handling Capacity.

- Power Handling capacity is limited due to dielectric breakdown.

Transmission Line Losses

□ Attenuation Loss.

- It happens due to the absorption of signal in the Transmission Line.
- Major Attenuation happens in dielectric material.

$$L_{att} = 10 \log \left[\frac{E_i - E_r}{E_t} \right]$$

$$\sim E \propto V^2 \Rightarrow E_i \propto V_i^2, E_r \propto V_r^2$$

$$E_t = [|V_i|^2 - |V_r|^2] e^{-2\alpha l}$$

$$\Rightarrow L_{att} = 10 \log \left[\frac{|V_i|^2 - |V_r|^2}{[|V_i|^2 - |V_r|^2] e^{-2\alpha l}} \right]$$

$$\Rightarrow L_{att} = 10 \log [e^{2\alpha l}]$$

$$\Rightarrow L_{att} = 20 \alpha l \log e$$

$$\Rightarrow L_{att} = 8.686 \alpha l$$

□ Reflection Loss.

- Reflection Loss happens due to the mismatch of the Transmission Line.

$$L_{\text{Ref}} = 10 \log \left[\frac{E_i}{E_i - E_r} \right]$$

$$= 10 \log \left[\frac{|V_i|^2}{|V_i|^2 - |V_r|^2} \right]$$

$$= 10 \log \left[\frac{1}{1 - \left| \left(\frac{V_r}{V_i} \right)^2 \right|} \right]$$

⇒ Reflection Coefficient $r = \frac{V_r}{V_i}$

$$\Rightarrow L_{\text{Ref}} = 10 \log \left[\frac{1}{1 - |r|^2} \right]$$

□ Transmission Loss.

- Transmission Loss is a loss in the Transmission Line.

$$L_{\text{Trans}} = 10 \log \left[\frac{E_i}{E_t} \right]$$

$$= 10 \log \left[\frac{E_i}{E_i - E_r} \times \frac{E_i - E_r}{E_t} \right]$$

$$= 10 \log \left[\frac{E_i}{E_i - E_r} \right] + 10 \log \left[\frac{E_i - E_r}{E_t} \right]$$

$$= L_{\text{Ref}} + L_{\text{Att}}$$

$$L_{\text{Trans}} = 10 \log \left[\frac{1}{1 - |\Gamma|^2} \right] + 8.686 \alpha l$$

□ Return Loss.

- Return Loss happens at the port of the Transmission Line.
- It happens due to an Impedance mismatch at the port of the Transmission Line.

$$L_{\text{Ret}} = 10 \log \left[\frac{E_r}{E_i} \right]$$
$$= 10 \log \left[\frac{|V_r|^2}{|V_i|^2} \right]$$

⇒ Reflection Coefficient, $r = \frac{V_r}{V_i}$

$$\Rightarrow L_{\text{Ret}} = 10 \log |r|^2$$

$$\Rightarrow L_{\text{Ret}} = 20 \log |r|$$

□ Insertion Loss.

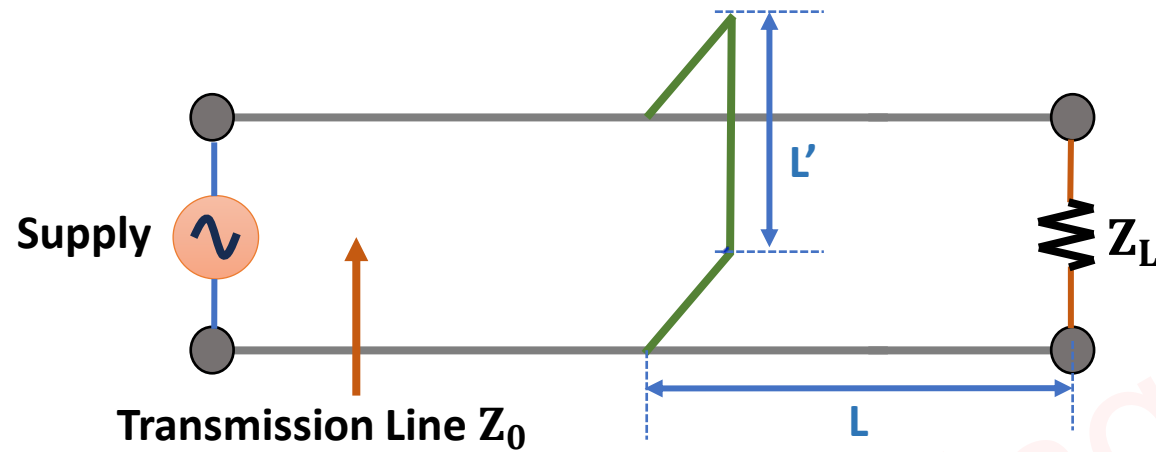
- Insertion Loss happens due to the insertion of the Transmission Line.

↪ E_1 is energy with Tx line

E_2 is energy without Tx line

$$\Rightarrow L_{ins} = 10 \log \left[\frac{E_2}{E_1} \right]$$

Impedance Matching using Single Stub



□ Basics of Impedance Matching

- For Maximum power transfer to the load Z_L , Impedance of Z_L and Z_0 should get matched, else there will be a reflection of signal.
- Reflection coefficient can be identified by:

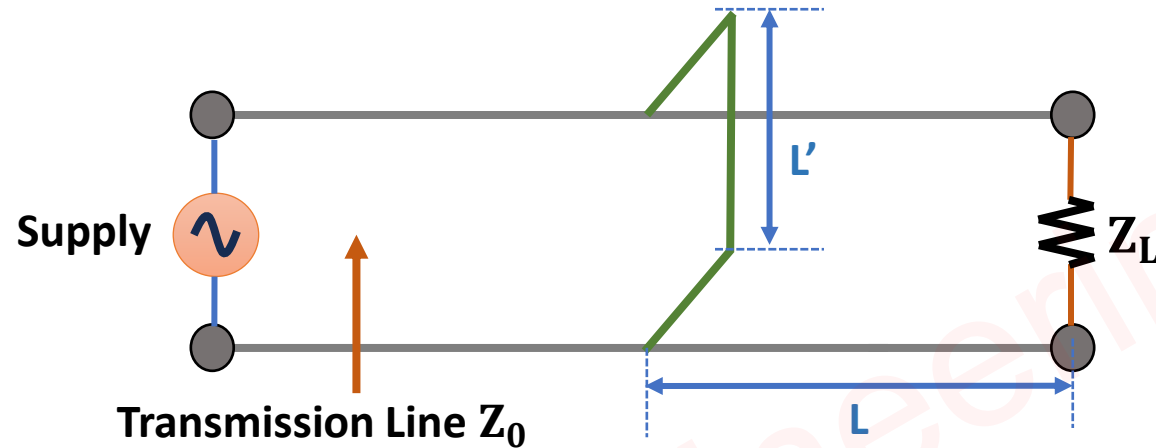
$$\Gamma = \frac{V_r}{V_i} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

- To minimize signal reflection, we can use a single stub (Short Circuited wire), which can provide impedance matching of Transmission line with load.

□ Basics of Single Stub Impedance Matching

- The stub (L) location should be such that Input Admittance should be $Y_1 = 1 + jX$.
- The Length of the Stub (L') should be such that the admittance of the stub should be $Y_2 = -jX$.
- So, Total Admittance is

$$Y = Y_1 + Y_2 = 1 \quad \rightarrow \quad Y = 1 = \frac{Y_L}{Y_0} \Rightarrow Y_L = Y_0 \Rightarrow Z_L = Z_0$$



⇒ Input Impedance

$$\Rightarrow Z_{in} = \frac{Z_L + j \tan \beta l}{1 + j Z_L \tan \beta l}$$

$$[Y = \frac{1}{Z}]$$

$$\Rightarrow \left(\frac{1}{Y_{in}} \right) = \frac{\left(\frac{1}{Y_L} \right) + j \tan \beta l}{1 + j \left(\frac{1}{Y_L} \right) \tan \beta l}$$

$$\Rightarrow Y_{in} = \frac{Y_L + j \tan \beta l}{1 + j Y_L \tan \beta l} \times \frac{1 - j Y_L \tan \beta l}{1 - j Y_L \tan \beta l}$$

$$\Rightarrow Y_{in} = \frac{Y_L + Y_L \tan^2 \beta l + j (\tan \beta l - Y_L^2 \tan \beta l)}{1 + Y_L^2 \tan^2 \beta l}$$

$$\Rightarrow Y_{in} = \frac{Y_L (1 + \tan^2 \beta l)}{1 + Y_L^2 \tan^2 \beta l} + j \frac{(1 - Y_L^2) \tan \beta l}{1 + Y_L^2 \tan^2 \beta l}$$

$\Rightarrow$ Here, Real Component = 1

$$\Rightarrow \frac{Y_L (1 + \tan^2 \beta l)}{1 + Y_L^2 \tan^2 \beta l} = 1$$

$$\Rightarrow Y_L + Y_L \tan^2 \beta l = 1 + Y_L^2 \tan^2 \beta l$$

$$\Rightarrow Y_L - 1 = (Y_L^2 - Y_L) \tan^2 \beta l$$

$$\Rightarrow \cancel{Y_L} - 1 = Y_L (\cancel{Y_L} - 1) \tan^2 \beta l$$

$$\Rightarrow \tan^2 \beta l = 1/Y_L$$

$$\Rightarrow \tan \beta l = \frac{1}{\sqrt{Y_L}}$$

$$\Rightarrow \tan \beta l = \sqrt{Z_L}$$

$$\Rightarrow \tan \beta l = \sqrt{\frac{Z_L}{Z_0}}$$

$$\Rightarrow \beta l = \tan^{-1} \sqrt{\frac{Z_L}{Z_0}} \quad \left[\because \beta = \frac{2\pi}{\lambda} \right]$$

$$\Rightarrow \boxed{l = \frac{\lambda}{2\pi} \tan^{-1} \sqrt{\frac{Z_L}{Z_0}}}$$

$\Rightarrow$ Here, we have short circuited stub,

$$\Rightarrow Z_{sc} = j Z_0 \tan \beta l'$$

$$\Rightarrow Z_{sc}/Z_0 = j \tan \beta l'$$

$$\Rightarrow Z_{sc} = j \tan \beta l'$$

$$\Rightarrow Y_{sc} = -j \cot \beta l'$$

$$\Rightarrow Y_{sc} = -jX, \text{ Here } X = \frac{(1-\gamma_L^2) \tan \beta l}{1 + \gamma_L^2 \tan^2 \beta l}$$

$$\Rightarrow \cot \beta l' = \frac{(1-\gamma_L^2) \tan \beta l}{1 + \gamma_L^2 \tan^2 \beta l}$$

$$\Rightarrow \cot \beta l' = \frac{(1-\gamma_L^2) (1/\sqrt{\gamma_L})}{1 + \gamma_L^2 (1/\cancel{\gamma_L})}$$

$$\Rightarrow \cot \beta l' = (1-\gamma_L) (1/\sqrt{\gamma_L})$$

$$\Rightarrow \cot \beta l' = \left(1 - \frac{Z_0}{Z_L}\right) \sqrt{\frac{Z_L}{Z_0}}$$

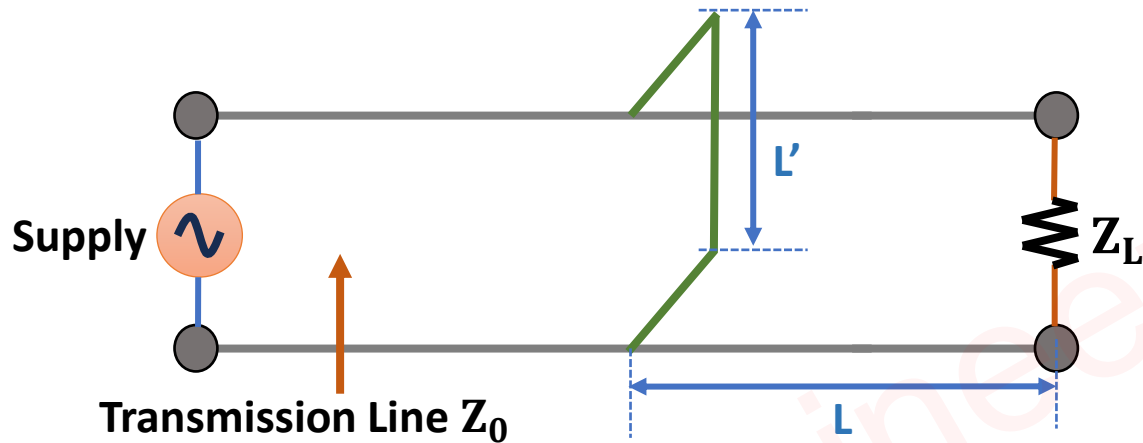
$$\Rightarrow \cot \beta l' = \frac{Z_L - Z_0}{\sqrt{Z_L Z_0}}$$

$$\Rightarrow \tan \beta l' = \frac{\sqrt{Z_L Z_0}}{Z_L - Z_0}$$

$$\Rightarrow \boxed{l' = \frac{\lambda}{2\pi} \tan^{-1} \left(\frac{\sqrt{Z_L Z_0}}{Z_L - Z_0} \right)}$$

Transmission Line Example on Impedance Matching using Single Stub

Question 1 – 100Ω line with air as a dielectric is terminated by load impedance 600Ω and is excited by 1GHz, with match generation. Find the position and length of the single stub.

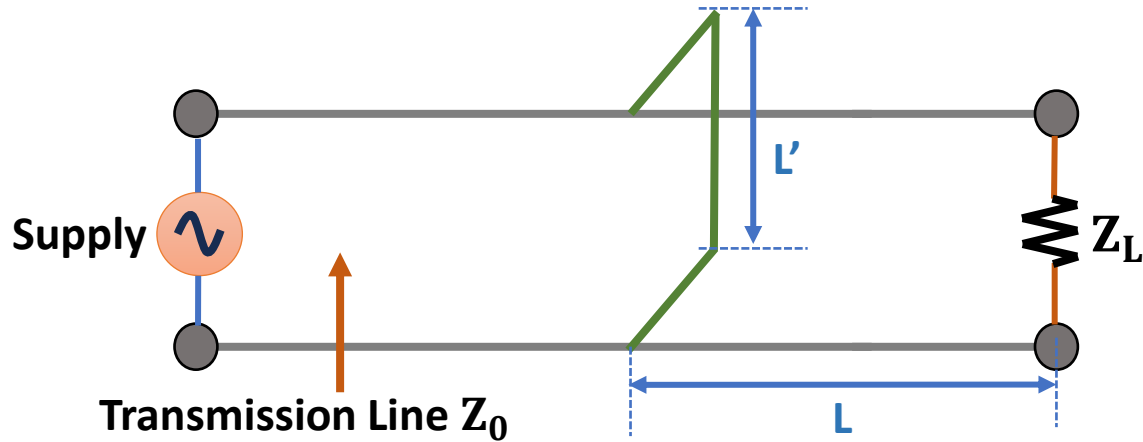


$$\begin{aligned} \Rightarrow Z_0 &= 100 \Omega \\ Z_L &= 600 \Omega \\ \lambda &= \frac{c}{f} = \frac{3 \times 10^8}{10^9} = 0.3 \text{ m} \end{aligned}$$

$$\Rightarrow L = \frac{\lambda}{2\pi} \tan^{-1} \sqrt{\frac{Z_L}{Z_0}} = \frac{0.3}{2\pi} \tan^{-1} \sqrt{\frac{600}{100}} = 0.056 \text{ m}$$

$$\Rightarrow L' = \frac{\lambda}{2\pi} \tan^{-1} \left(\frac{\sqrt{Z_L Z_0}}{Z_L - Z_0} \right) = \frac{0.3}{2\pi} \tan^{-1} \left(\frac{\sqrt{600 \times 100}}{500} \right) = 0.0217 \text{ m}$$

Question 2 – A load $(50 + j100)\Omega$ is connected across a 50Ω . Design a short-circuited stub to provide impedance matching at 30MHz.



$$\Rightarrow Z_L = 50 + j100 \Omega$$

$$Z_0 = 50 \Omega$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{30 \times 10^6} = 10 \text{ m}$$

$$\Rightarrow r = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$= \frac{50 + j100 - 50}{50 + j100 + 50}$$

$$= \frac{j100}{100 + j100} = \frac{100 \angle 90}{141.4 \angle 45}$$

$$= 0.707 \angle 45$$

$$\leadsto |r| = 0.707, \phi = 45^\circ$$

$$\begin{aligned} \gamma \quad L &= \frac{\lambda}{4\pi} [\phi + \pi - \cos^{-1}|r|] \\ &= \frac{10}{4\pi} \left[\frac{\pi}{4} + \pi - \cos^{-1} 0.707 \right] \\ &= \frac{10}{4\pi} [\cancel{\pi/4} + \pi - \cancel{\pi/4}] \\ &= 2.5 \text{ m} \end{aligned}$$

$$\begin{aligned} \leadsto L' &= \frac{\lambda}{2\pi} \tan^{-1} \left(\frac{\sqrt{1+|r|^2}}{2|r|} \right) \\ &= \frac{10}{2\pi} \tan^{-1} \left(\frac{\sqrt{1+0.707^2}}{2 \times 0.707} \right) \\ &= 1.13 \text{ m} \end{aligned}$$

Transmission Line Example on Losses of Transmission Line

Question 1 – A 50Ω generator feeds a 600Ω Lossless Transmission line. If the line is 200m long and terminated by the load of 500Ω , Determine in dBs

- Reflection Loss
- Transmission Loss
- Return Loss

$$\begin{aligned}\Rightarrow Z_g &= 50\Omega \\ \alpha &= 0 \\ Z_0 &= 600\Omega \\ l &= 200\text{m} \\ Z_L &= 500\Omega\end{aligned}$$

$$\begin{aligned}\Rightarrow \Gamma &= \frac{Z_L - Z_0}{Z_L + Z_0} \\ &= \frac{500 - 600}{500 + 600} \\ &= \frac{-100}{1100} \\ &= -0.09\end{aligned}$$

$$\begin{aligned}\Rightarrow L_{\text{ref}} &= 10 \log \left[\frac{1}{1 - |\Gamma|^2} \right] \\ &= 10 \log \left[\frac{1}{1 - 0.09^2} \right] \\ &= 0.035 \text{ dB}\end{aligned}$$

$$\begin{aligned}\leadsto L_{\text{Trans}} &= L_{\text{Att}} + L_{\text{Ref}} \\ &= 8.686 \text{ dB} + 10 \log \left[\frac{1}{1-|r_1|^2} \right] \\ &= 0.035 \text{ dB}\end{aligned}$$

$$\begin{aligned}\leadsto L_{\text{Ref}} &= 20 \log |r_1| \\ &= 20 \log 0.09 \\ &= -20.91 \text{ dB}\end{aligned}$$

Transmission Line Examples

Question 1 – For a Transmission Line, RLGC Components are given by 5Ω , 1mH , 0.1S , and $1\mu\text{F}$, respectively. Find Z_0 , γ , α , and β at 1MHz .

$$\Rightarrow R = 5\Omega$$

$$L = 1\text{mH}$$

$$G = 0.1\text{S}$$

$$C = 1\mu\text{F}$$

$$f = 1\text{MHz}$$

$$\Rightarrow Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$= \sqrt{\frac{5 + j2\pi \times 10^6 \times 10^{-3}}{0.1 + j2\pi \times 10^6 \times 10^{-6}}}$$

$$= \sqrt{\frac{6.28 \times 10^3 \angle 89.95^\circ}{6.28 \angle 89.08^\circ}}$$

$$= 31.62 \angle 0.44^\circ$$

$$\Rightarrow \gamma = \alpha + j\beta$$

$$= \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$= \sqrt{6.28 \times 10^3 \times 6.28 \times 10^{-6} \angle (89.95^\circ + 89.08^\circ)}$$

$$= 198 \angle 89.52^\circ$$

$$= \frac{1.65}{\alpha} + j \frac{198}{\beta}$$

Question 2 – A Lossless Transmission Line of 50Ω is having $V_{\text{MAX}} = 2.5\text{V}$, $V_{\text{MIN}} = 1\text{V}$. When terminated by unknown load. Find VSWR, Reflection Coefficient and Load Impedance.

$$\Rightarrow Z_0 = 50\Omega$$

$$V_{\text{max}} = 2.5\text{V}$$

$$V_{\text{min}} = 1\text{V}$$

$$\Rightarrow \text{VSWR} = \frac{V_{\text{max}}}{V_{\text{min}}} = \frac{2.5}{1} = 2.5$$

$$\Rightarrow \Gamma = \frac{\text{VSWR} - 1}{\text{VSWR} + 1} = \frac{2.5 - 1}{2.5 + 1} = \frac{1.5}{3.5} = 0.428$$

$$\begin{aligned}\Rightarrow \text{VSWR} &= \frac{Z_L}{Z_0} \quad (Z_L > Z_0) \\ &= \frac{Z_0}{Z_L} \quad (Z_0 > Z_L)\end{aligned}$$

$$\Rightarrow 2.5 = \frac{Z_L}{50} \Rightarrow \boxed{Z_L = 125\Omega}$$

$$\Rightarrow \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Rightarrow 0.428 = \frac{Z_L - 50}{Z_L + 50}$$

$$\Rightarrow 0.428 Z_L + 50 \times 0.428 = Z_L - 50$$

$$\Rightarrow Z_L - 0.428 Z_L = 50 \times 0.428 + 50$$

$$\Rightarrow \boxed{Z_L = 125\Omega}$$

Transmission Line Examples

Question 3 – The primary line constants of a Transmission Line per km are specified as $R=42.9\Omega$, $L=0.7\text{mH}$, $G=0.1\mu\text{S}$, and $C=0.1\mu\text{F}$, respectively. Find Z_0 , γ , α , β and v_p at 5000rad/sec .

$$\Rightarrow R = 42.9 \Omega$$

$$L = 0.7 \text{ mH}$$

$$G = 0.1 \mu\text{S}$$

$$C = 0.1 \mu\text{F}$$

$$\omega = 5000 \text{ rad/sec}$$

$$\Rightarrow Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

$$= \sqrt{\frac{42.9 + j5000 \times 0.7 \times 10^{-3}}{10^{-7} + j5000 \times 10^{-7}}}$$

$$= \sqrt{\frac{43.04 \angle 4.66}{5 \times 10^{-4} \angle 89.72}}$$

$$= 293.39 \angle -42.53 \Omega$$

$$\Rightarrow \gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$$

$$= \sqrt{43.04 \times 5 \times 10^{-4} \angle (4.66 + 89.72)}$$

$$= 0.14 \angle 47.19$$

$$= \frac{0.095}{\alpha} + j \frac{0.1027}{\beta}$$

$$[Np/km] \quad [Rad/km]$$

$$\sim v_p = \frac{\omega}{\beta} \frac{[Rad/Sec]}{[Rad/km]} = [km/sec]$$

$$= \frac{5000}{0.1027}$$

$$= 48.68 \times 10^3 \text{ km/sec}$$

Question 4 – A Transmission Line has the following parameters:

$R=2\Omega/\text{m}$, $L=8\text{nH}/\text{m}$, $G=0.5\text{mS}/\text{m}$, $C=0.23\text{pF}/\text{m}$, and $f=1\text{GHz}$

Find Z_0 , γ , α , β and v_p .

$$\Rightarrow R = 2 \Omega/\text{m}$$

$$L = 8 \text{ nH}/\text{m}$$

$$G = 0.5 \text{ mS}/\text{m}$$

$$C = 0.23 \text{ pF}/\text{m}$$

$$\begin{aligned}\rightarrow Z_0 &= \sqrt{\frac{R + j\omega L}{G + j\omega C}} \\&= \sqrt{\frac{2 + j 2 \times 3.14 \times 10^9 \times 8 \times 10^{-9}}{0.5 \times 10^{-3} + j 2 \times 3.14 \times 10^9 \times 0.23 \times 10^{-12}}} \\&= \sqrt{\frac{50.27 \angle 87.72}{1.53 \times 10^{-3} \angle 70.91}} \\&= 181.26 \angle 8.4 \Omega\end{aligned}$$

$$\begin{aligned}\Rightarrow \gamma &= \sqrt{(R + j\omega L)(G + j\omega C)} \\&= \sqrt{50.27 \times 1.53 \times 10^{-3} \angle (87.72 + 70.91)}\end{aligned}$$

$$= 0.2773 \angle 79.31$$

$$= \frac{0.0514}{\alpha} + j \frac{0.272}{\beta}$$

$$[Np/m] \quad [Rad/m]$$

$$\Rightarrow v_p = \frac{\omega}{\beta} \frac{[Rad/sec]}{[Rad/m]} = m/sec$$

$$= \frac{2\pi f}{\beta}$$

$$= \frac{2\pi \times 10^9}{0.272}$$

$$= 2.308 \times 10^{10} m/sec$$

